Verification 08_M_Simulator_FEM_2D

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1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the analytical solutions for 2D linear elasticity.

2 Governing Equation

The mechanical equilibrium equation is

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \rho \ddot{\mathbf{u}}, \tag{1}$$

where $\boldsymbol{\sigma}$ is the Cauchy stress tensor, $\mathbf{b}$ is the body force per unit volume, ρ is the material density, and $\mathbf{u}$ is the displacement vector. For the quasi-static problems considered here, inertial effects are neglected, giving

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0}. \tag{2}$$

For an isotropic linear elastic material, the stress–strain relationship is

$$\begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \tau_{xy} \end{bmatrix} = \mathbf{D} \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix}, \tag{3}$$

where the constitutive matrix $\mathbf{D}$ depends on whether a plane stress or plane strain assumption is adopted. For plane stress,

$$\mathbf{D} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix}, \tag{4}$$

whereas for plane strain,

$$\mathbf{D} = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \dfrac{1-2\nu}{2} \end{bmatrix}. \tag{5}$$

3 Simulation Parameters

The simulation parameters are given in Tables 1, 2, and 3. Note that compression is positive.

Parameter	Symbol	Value
Number of elements	N_x	81
Number of elements	N_y	81
Domain length	L_x	30 m
Domain width	L_y	10 m
Young's Modulus	E	1×10^9 Pa
Poisson's ratio	ν	0.3
Load on top edge	S_y	10 Pa

Table 1: Simulation parameters for first test

Parameter	Symbol	Value
Number of elements	N_x	81
Number of elements	N_y	81
Domain length	L_x	30 m
Domain width	L_y	10 m
Young's Modulus	E	1×10^9 Pa
Poisson's ratio	ν	0.3
Load on right edge	S_x	10 Pa

Table 2: Simulation parameters for second test

4 Results

4.1 Test 1

In this test, a vertical load of 10 Pa is applied to the top boundary, Table 1. The bottom boundary is fixed in the vertical direction. The left boundary is fixed in the horizontal direction. The expected vertical stress is 10 Pa throughout the block. This is recovered, Figures 1 and 2. The remaining stress components are zero to numerical precision.

This simulation is in plane strain. The expected vertical strain in this case is $\epsilon_{yy} = \frac{1-\nu^2}{E} S_y = 9.1 \times 10^{-9}$. This result is recovered, Figures 3 and 4.

The expected horizontal strain in this case is $\epsilon_{xx} = \frac{-\nu(1+\nu)}{E} S_y = -3.9 \times 10^{-9}$. This result is recovered, Figures 5 and 6.

The volumetric strain in this case is $\epsilon_v = \frac{(1+\nu)(1-2\nu)}{E} S_y = 5.2 \times 10^{-9}$. This result is recovered, Figures 7 and 8.

4.2 Test 2

A second similar test is performed with the load now being on the right-hand edge, Table 2. This time the simulation is performed in plane stress. Here, the expected horizontal stress is 10 Pa. This result is recovered, Figures 9 and 10.

In this case, the horizontal strain is expected to be $\epsilon_{xx} = S_x/E = 1.0 \times 10^{-8}$. This result is recovered, Figures 11 and 12.

The vertical strain is expected to be $\epsilon_{yy} = -\nu S_x/E = -3.0 \times 10^{-9}$. This result is recovered, Figures 13 and 14.

The volumetric strain in this case is $\epsilon_v = \frac{1-\nu}{E} S_x = 7.0 \times 10^{-9}$. This result is recovered, Figures 15 and 16.

4.3 Test 3

In this case a uniform shear stress is applied to the top edge. Specifically, a stress of -10 Pa is applied in the x-direction on the top nodes, Table 3. The bottom nodes are fixed in displacement in both the x and y directions. The simulation is plane strain. The expected shear stress is 10 Pa in this case. This result is recovered, Figures 17 and 18.

The shear modulus is found as $G = \frac{E}{2(1+\nu)} = 3.8462 \times 10^8$ Pa. The engineering shear strain is expected to be $\gamma_{xy} = \frac{\tau}{G} = 2.6 \times 10^{-8}$. This result is recovered, Figures 19 and 20.

Parameter	Symbol	Value
Number of elements	N_x	81
Number of elements	N_y	81
Domain length	L_x	30 m
Domain width	L_y	10 m
Young's Modulus	E	1×10^9 Pa
Poisson's ratio	ν	0.3
Load on top edge	S_x	-10 Pa

Table 3: Simulation parameters for third test

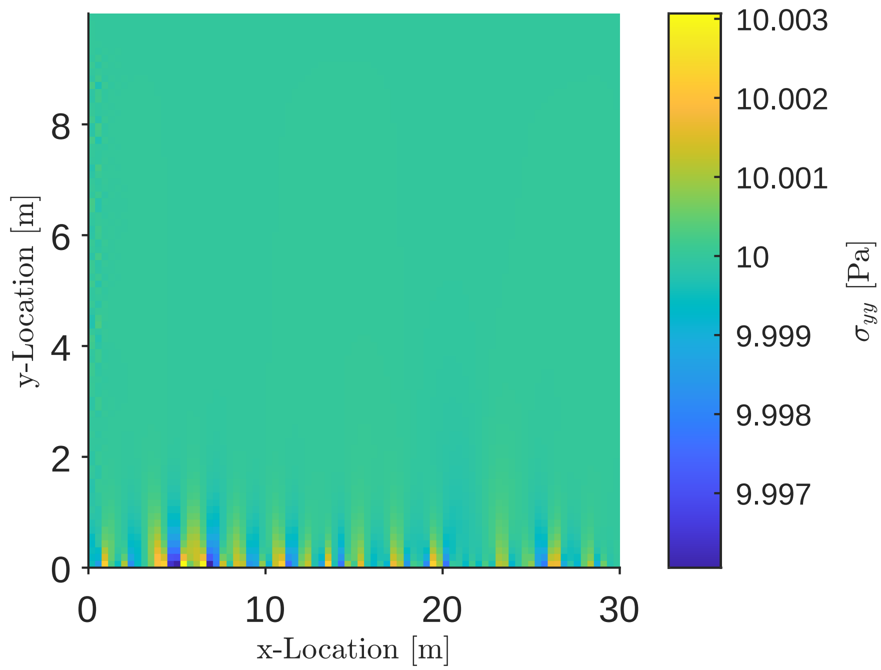

Figure 1: Vertical stress found in MATLAB for a vertical stress of 10 Pa applied to the top edge.

5 Discussion

The numerical stress and strain fields reproduce the analytical solutions for all three benchmark problems. Uniform normal loading in both plane strain and plane stress, together with uniform shear loading, are recovered to machine precision. These tests verify the implementation of the constitutive law, equilibrium equations, boundary conditions, and strain computation.

6 Conclusion

These benchmark problems verify the implementation of the stiffness matrix assembly, constitutive relations, boundary conditions, and stress and strain recovery procedures for two-dimensional linear elasticity.

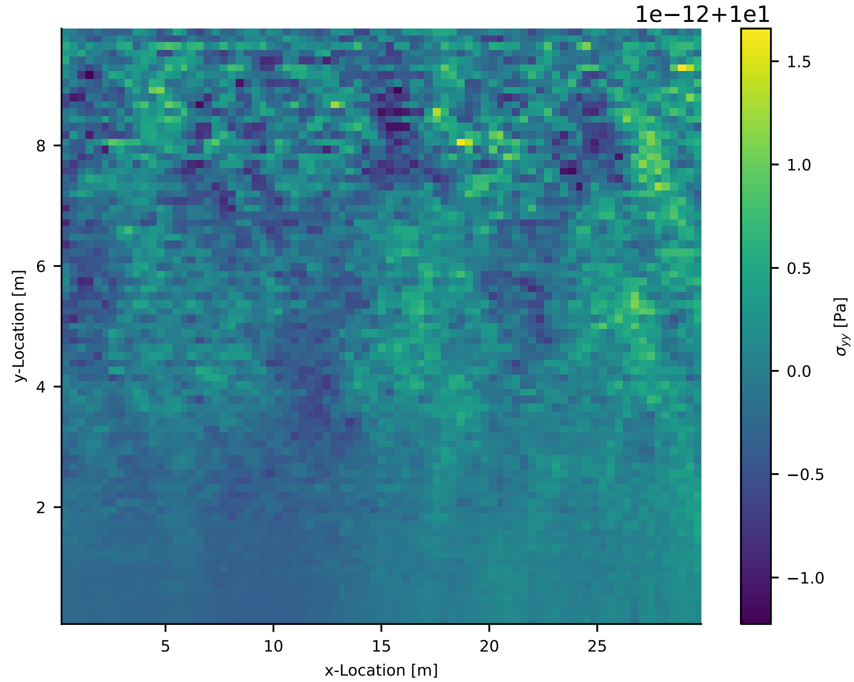

Figure 2: Vertical stress found in Python for a vertical stress of 10 Pa applied to the top edge.

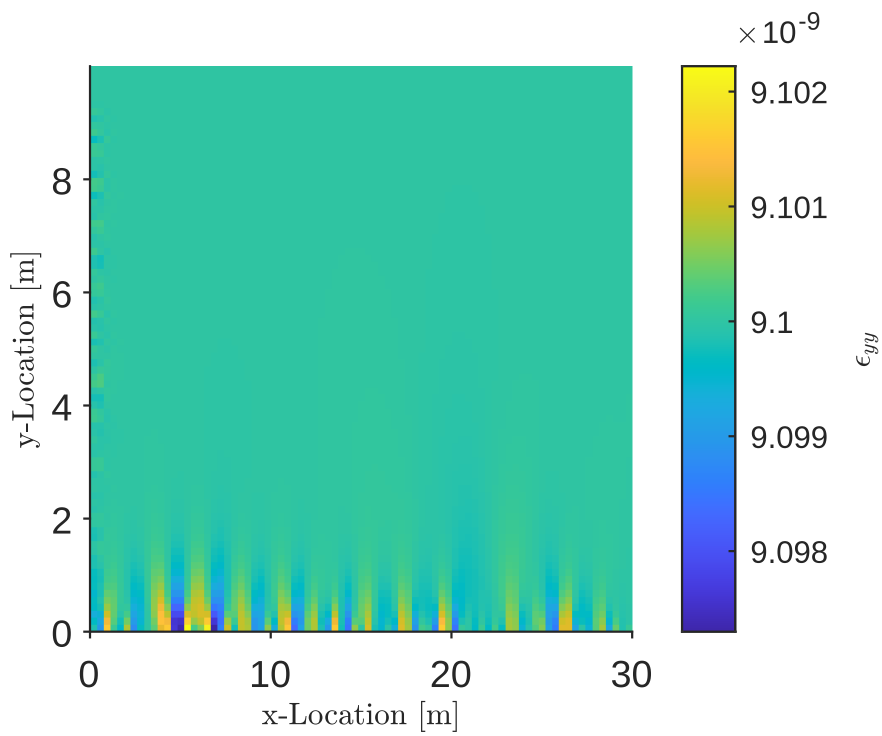

Figure 3: Vertical strain found in MATLAB for a vertical stress of 10 Pa applied to the top edge.

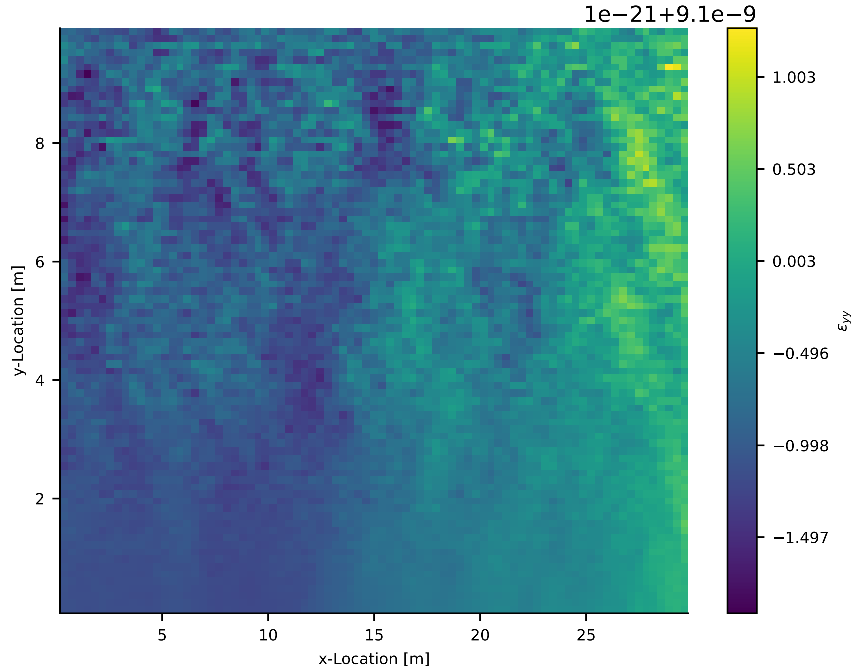

Figure 4: Vertical strain found in Python for a vertical stress of 10 Pa applied to the top edge.

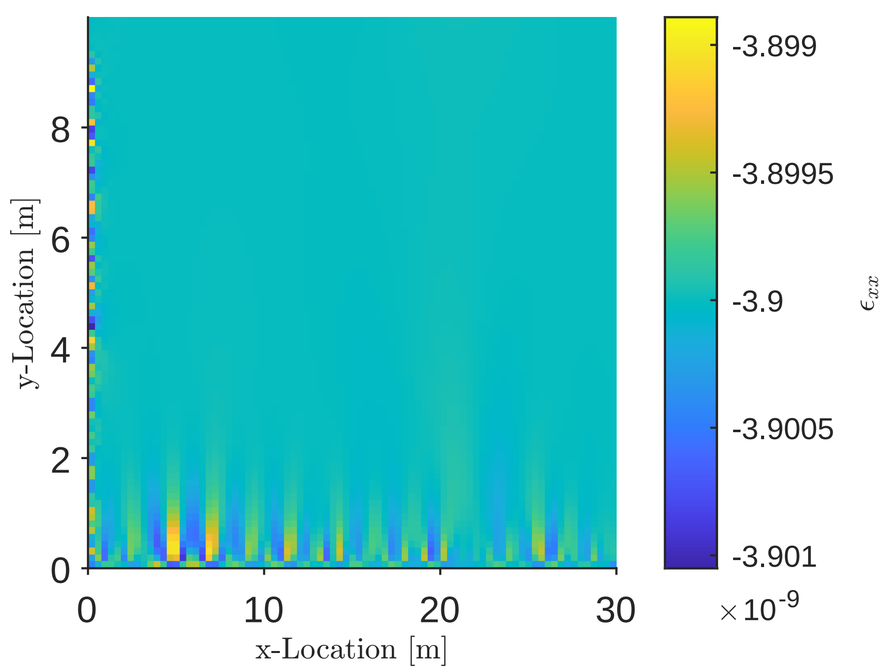

Figure 5: Horizontal strain found in MATLAB for a vertical stress of 10 Pa applied to the top edge.

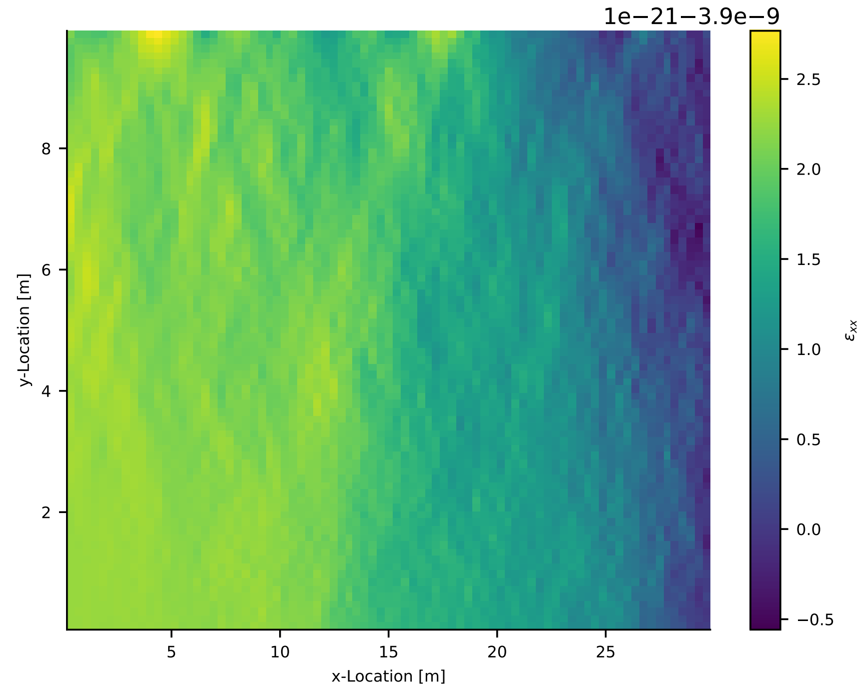

Figure 6: Horizontal strain found in Python for a vertical stress of 10 Pa applied to the top edge.

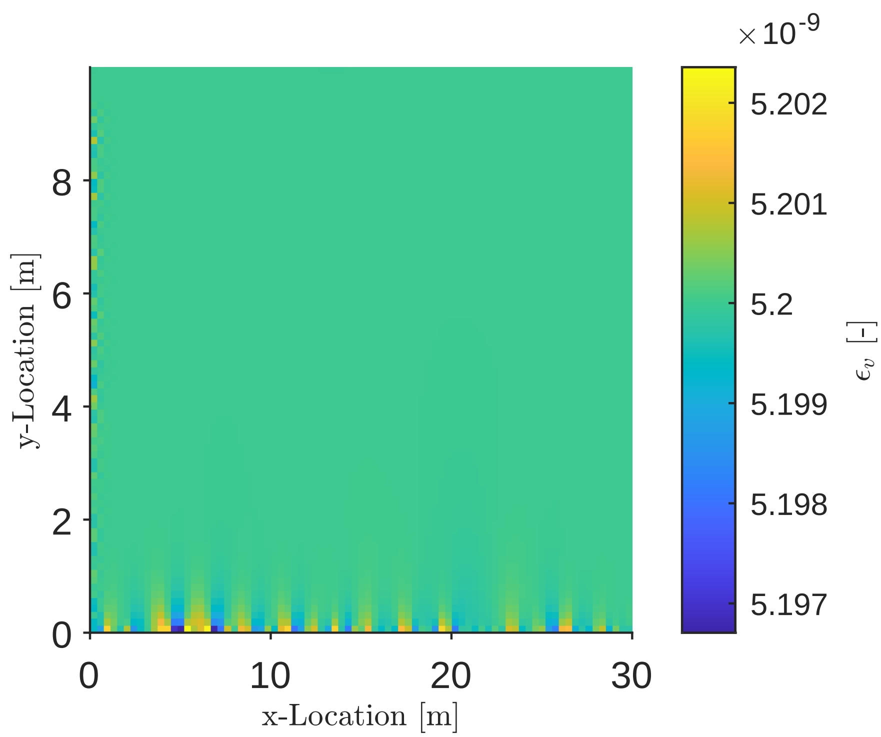

Figure 7: Volumetric strain found in MATLAB for a vertical stress of 10 Pa applied to the top edge.

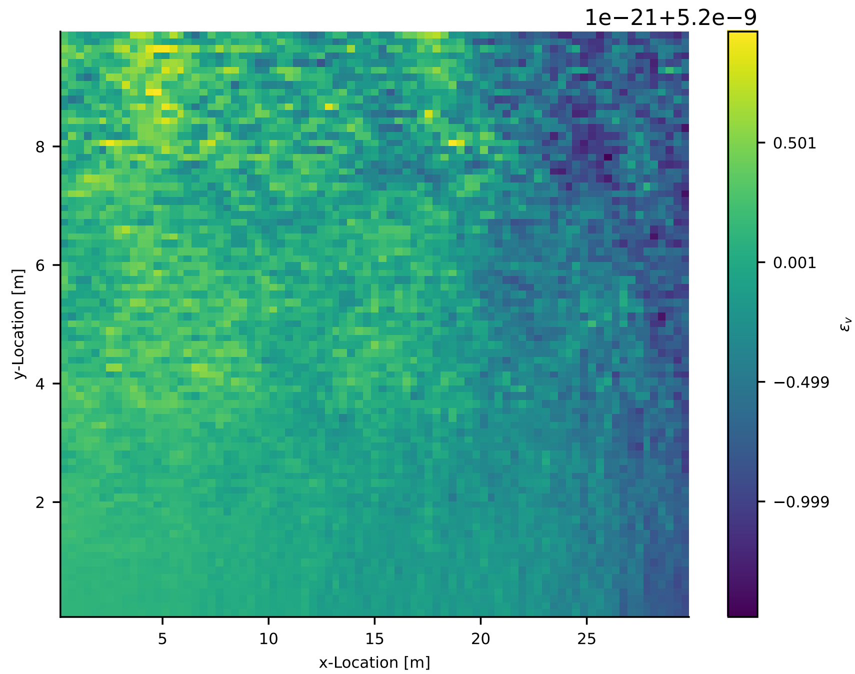

Figure 8: Volumetric strain found in Python for a vertical stress of 10 Pa applied to the top edge.

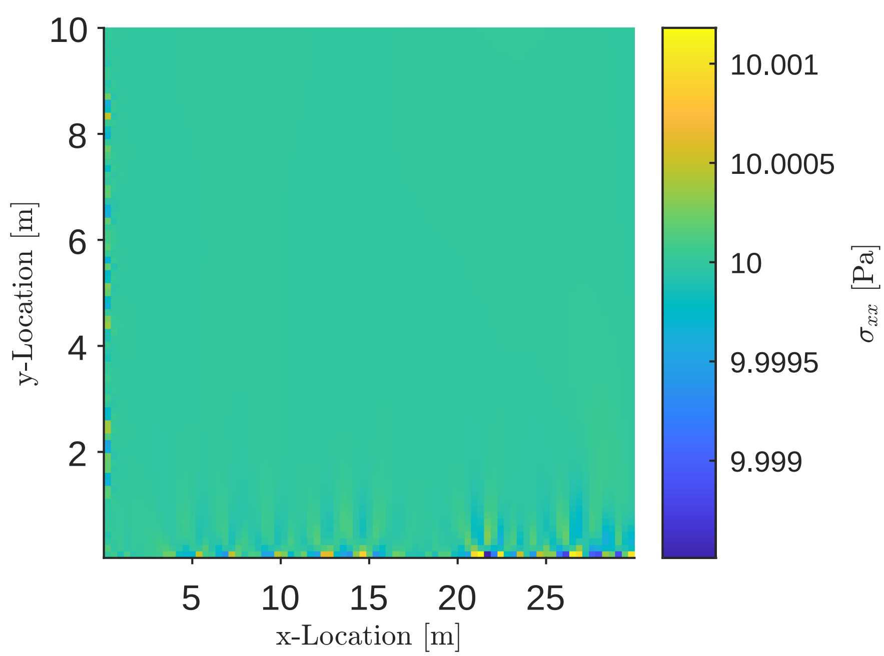

Figure 9: Horizontal stress found in MATLAB for a horizontal stress of 10 Pa applied to the right edge.

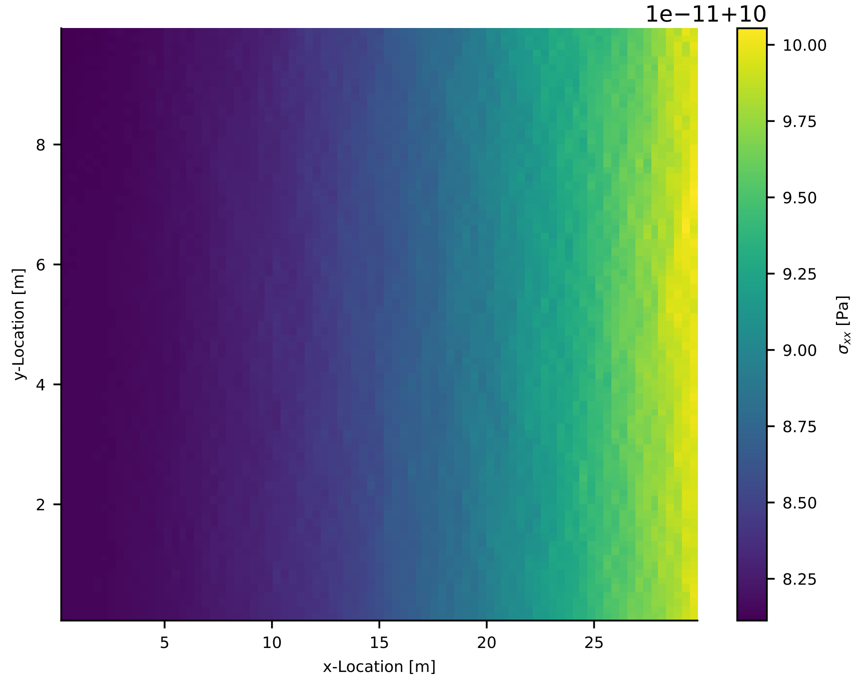

Figure 10: Horizontal stress found in Python for a horizontal stress of 10 Pa applied to the right edge.

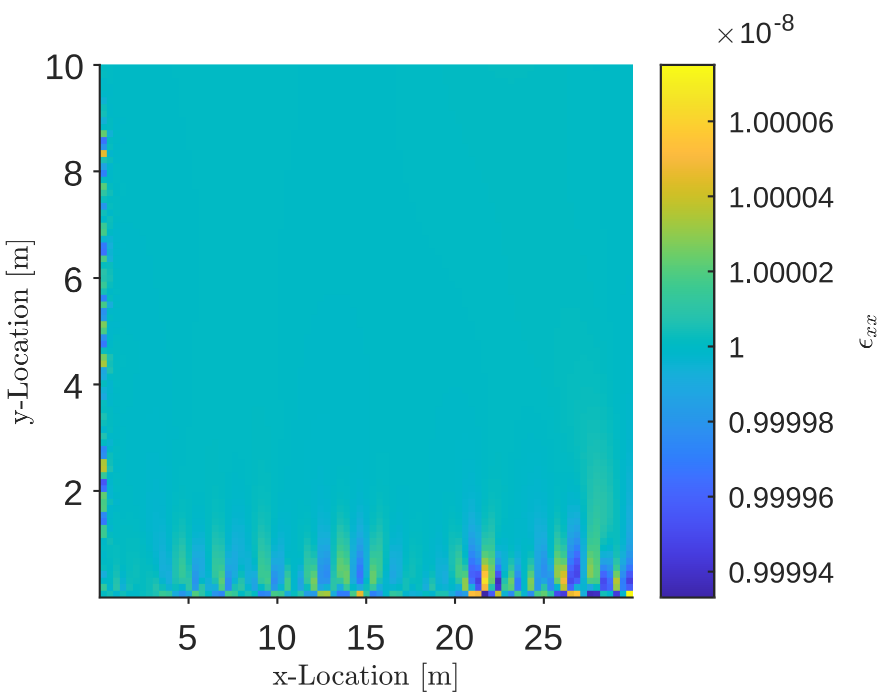

Figure 11: Horizontal strain found in MATLAB for a horizontal stress of 10 Pa applied to the right edge.

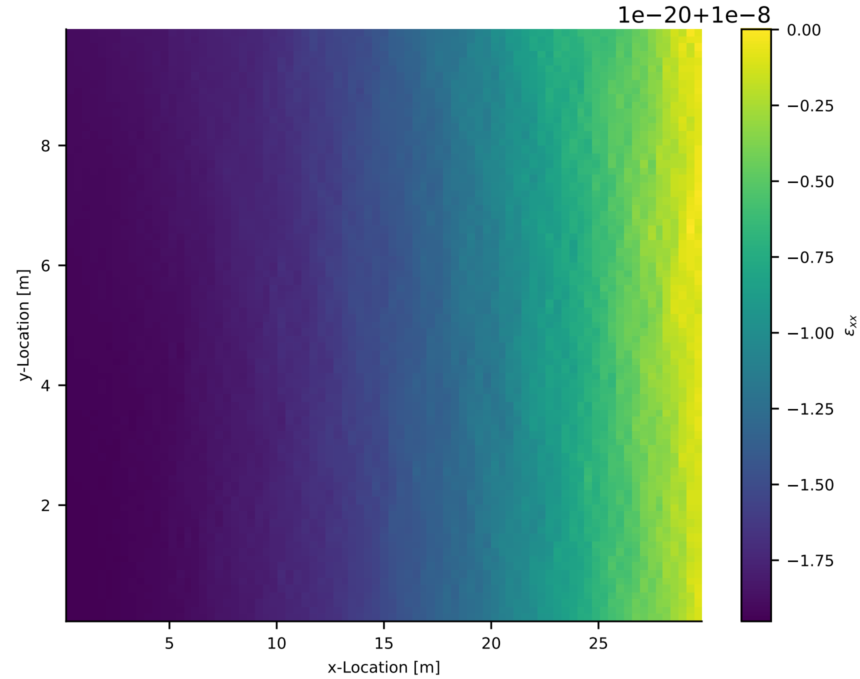

Figure 12: Horizontal strain found in Python for a horizontal stress of 10 Pa applied to the right edge.

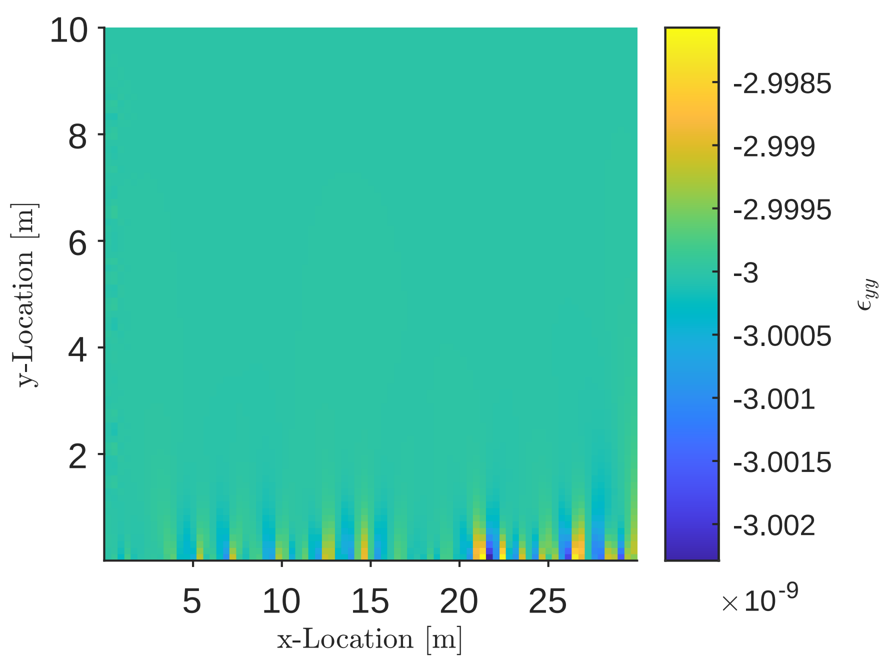

Figure 13: Vertical strain found in MATLAB for a horizontal stress of 10 Pa applied to the right edge.

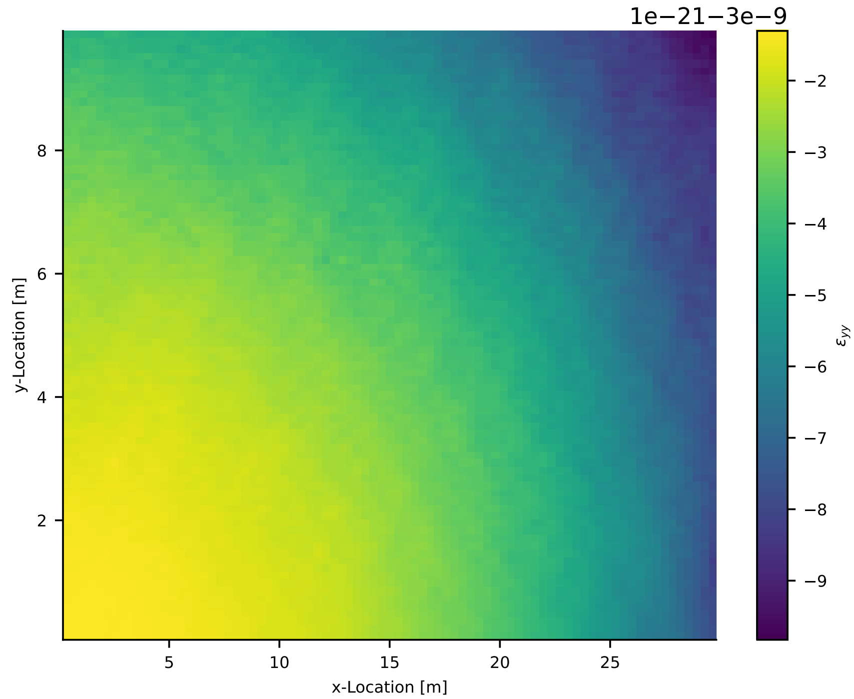

Figure 14: Vertical strain found in Python for a horizontal stress of 10 Pa applied to the right edge.

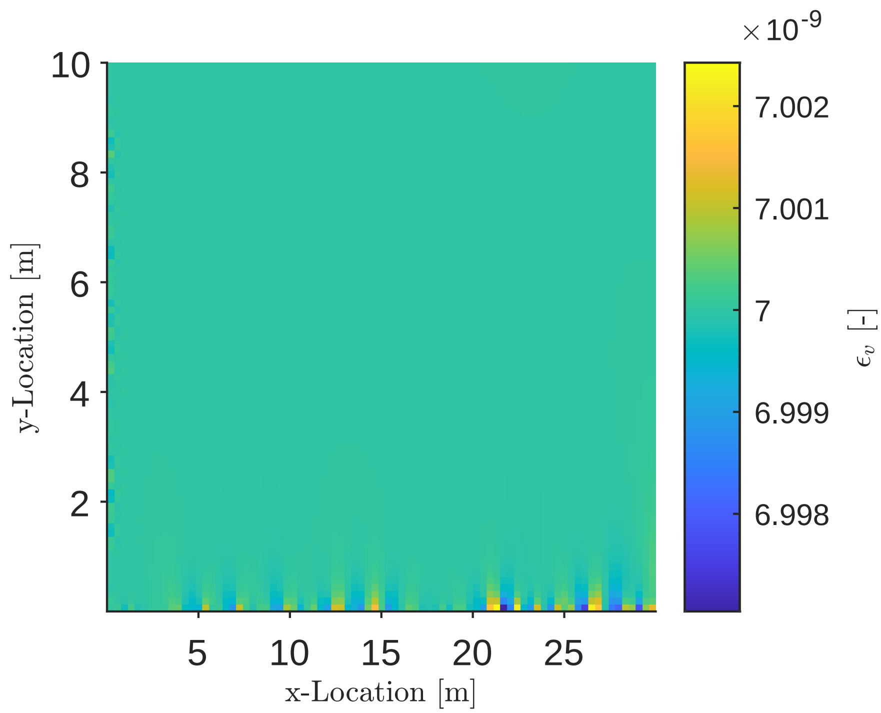

Figure 15: Volumetric strain found in MATLAB for a horizontal stress of 10 Pa applied to the right edge.

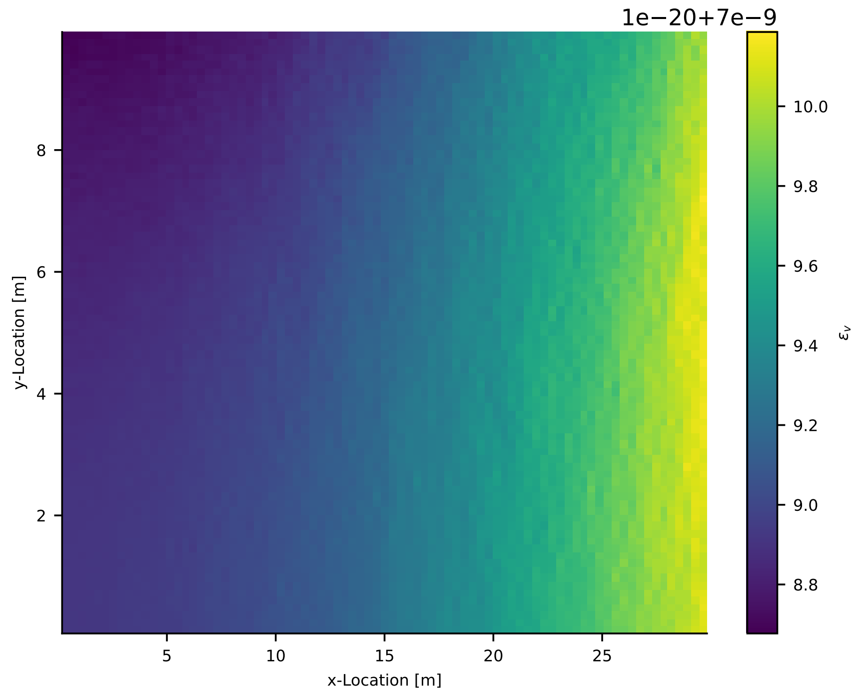

Figure 16: Volumetric strain found in Python for a horizontal stress of 10 Pa applied to the right edge.

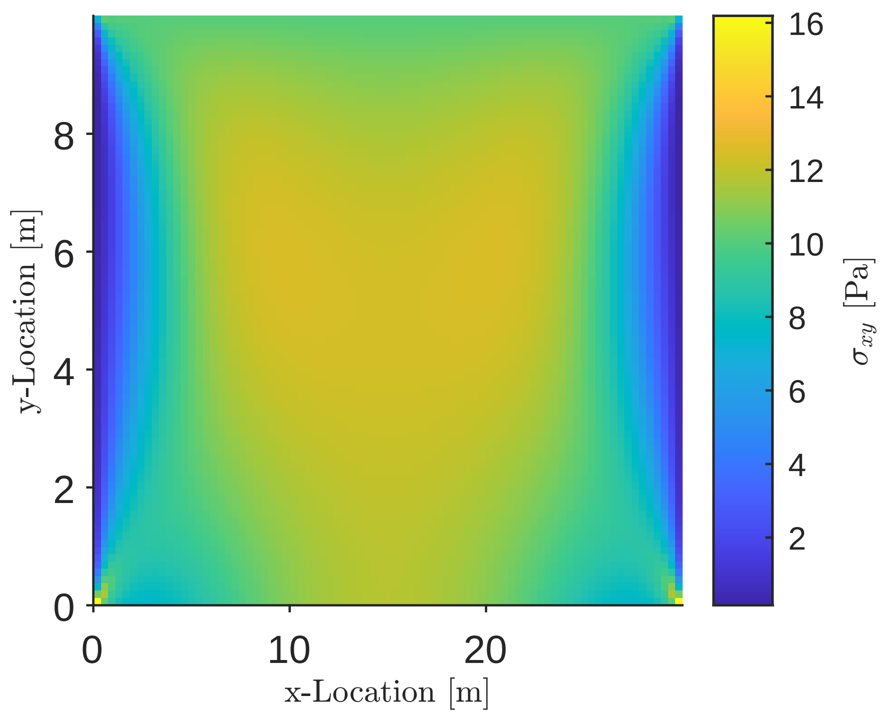

Figure 17: Shear stress found in MATLAB for a horizontal stress of -10 Pa applied to the top edge.

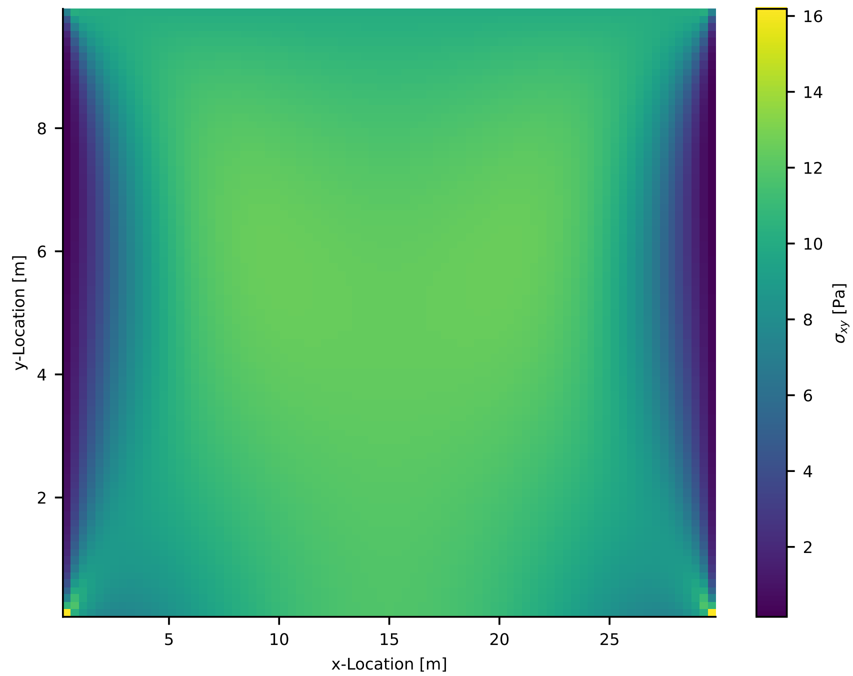

Figure 18: Shear stress found in Python for a horizontal stress of -10 Pa applied to the top edge.

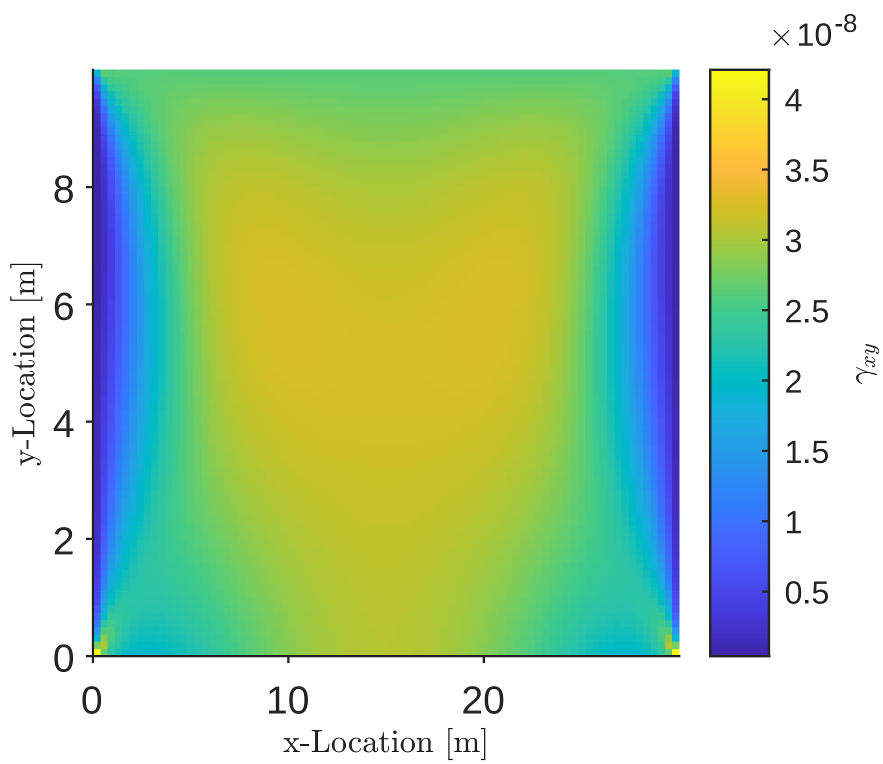

Figure 19: Engineering shear strain found in MATLAB for a horizontal stress of -10 Pa applied to the top edge.

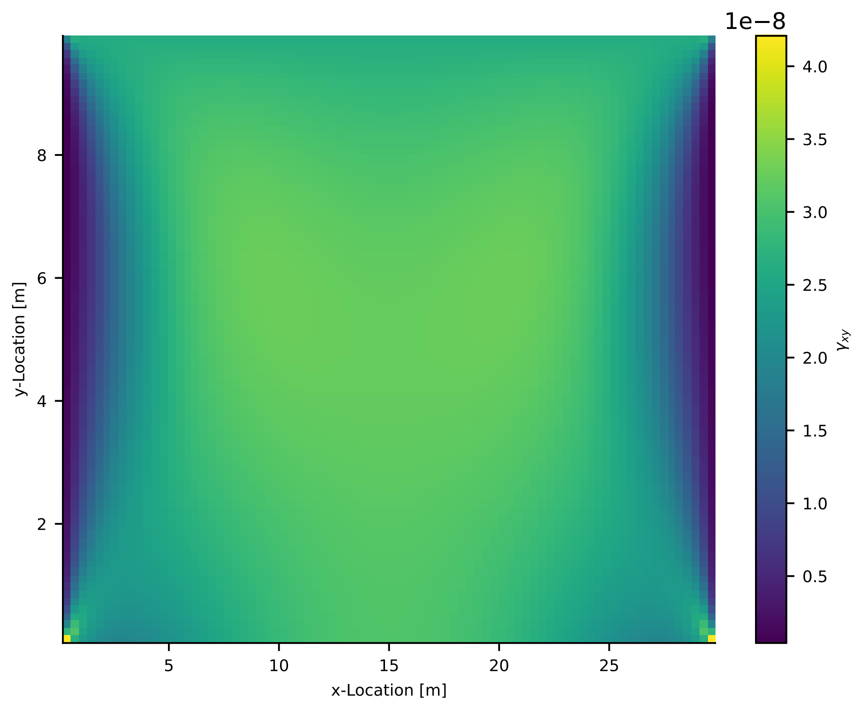

Figure 20: Engineering shear strain found in Python for a horizontal stress of -10 Pa applied to the top edge.